MAL Omics: Proteomics Analysis Report

COPDGene, SomaScan (∼4,979 proteins), $n = 1{,}306$

Study. COPDGene is a longitudinal cohort study of current and former smokers enrolled at 21 US clinical centers. Subjects were seen at Phase 2 (∼2013) and Phase 3 (∼2019). Blood was drawn at Phase 2 and profiled by SomaScan, an aptamer-based platform that quantifies ∼4,979 plasma proteins simultaneously as RFU (Relative Fluorescence Units). This analysis uses $n = 1{,}306$ subjects with complete proteomics, phenotype, and MAL data.

MAL (Mechanically Affected Lung). Computed tomography (CT)-derived continuous score (range 0.01–0.40) measuring the fraction of lung volume under destructive mechanical stress. Higher MAL = more lung tissue under stress. MAL is computed from inspiratory/expiratory CT image registration and is distinct from standard emphysema measures (e.g., %LAA-950). Variable name in data: `meanmal`.

MAL is itself a strong predictor of who will develop worse emphysema over time: higher mechanical stress on lung tissue drives progressive structural damage. However, MAL requires specialized CT analysis and is not available from a routine blood draw.

Goal. (1) Identify which Phase 2 plasma proteins are associated with MAL, establishing that mechanical lung stress leaves a detectable signal in blood. (2) Use those proteins to build a Proteomic Risk Score (proRS) from a Phase 2 blood draw that approximates MAL and can predict who will develop worse emphysema by Phase 3. If successful, the proRS would allow MAL-like risk stratification without requiring CT image registration. The proteins themselves are not measured at Phase 3; they serve as baseline predictors of the CT density change that occurs between visits (∼6 years apart).

Outcome. Emphysema progression = change in CT lung density Phase 3 − Phase 2 (`Change_lung_density_vnb_P2_P3`, Hounsfield Units [HU]). Negative = worsened. Mean ≈ 1 HU (standard deviation [SD] 7) in this cohort; a negative value means the lung became less dense (more emphysematous) between visits.

Data Sources and Variable Names

Four CSV files were used. All joins are on `sid`.

`data/csv_exports/prot.csv`: SomaScan proteomics

Column	Description
`sid`	Subject identifier (join key)
`X{seqID}`	Raw RFU value for each protein (e.g. `X10000.28`). ∼4,979 columns matching pattern `^X[0-9]`. Renamed to `log2_X{seqID}` after $\log_2$ transform.

data/csv_exports/pheno_MAL.csv: Phenotypes and covariates

Column	Used as
sid	Join key
Change_lung_density_vnb_P2_P3	Outcome: emphysema progression (HU)
lung_density_vnb_P2	Baseline lung density (HU)
Age_P2	Age (years)
gender	Sex (coded 1/2)
race	Race (coded 1/2)
ATS_PackYears_P2	Pack-years of smoking
SmokCigNow_P2	Current smoking (0/1)
scanner_model_clean_P2	CT scanner model
FEV1_FVC_post_P2	FEV_1 (forced expiratory volume in 1 second) / FVC (forced vital capacity)
PC1 – PC5	Ancestry principal components
fev1_pp_gli_global_P2	FEV_1% predicted (GLI global reference)
BMI_P2	Body mass index

data/Data/MAL.csv: MAL scores

Column	Description
sid	Subject identifier (join key)
meanmal	MAL score (continuous, range 0.01–0.40)

data/csv_exports/CellData.csv: Phase 2 BMI

Column	Used as	Note
sid	Join key	
Phase_study	Phase filter	Rows with Phase_study == 2 used
BMI	BMI_P2 in analysis	All 1,306 analysis subjects present, no missingness

data/csv_exports/metadat5k.csv: Protein annotation

Column	Description
seqID	Protein sequence identifier; joins to log2_X{seqID}
EntrezGeneSymbol	HGNC (HUGO Gene Nomenclature Committee) gene symbol (e.g. ADIPOQ)
Target	Short protein name (e.g. Adiponectin)
TargetFullName	Full protein name

Join logic. Three-way inner join: prot $\bowtie_{\texttt{sid}}$ pheno $\bowtie_{\texttt{sid}}$ MAL. Metadata joined on `seqID = sub("^log2_X", "", predictor)`.

File sizes: prot.csv has 5,670 subjects; MAL.csv has 5,024 subjects; pheno_MAL.csv has 10,652 subjects. After the 3-way inner join on sid: $n = 4{,}630$.

Complete-case filter then requires non-missing values for all of: Change_lung_density_vnb_P2_P3 (Phase 3 outcome), lung_density_vnb_P2, meanmal, Age_P2, gender, race, ATS_PackYears_P2, SmokCigNow_P2, scanner_model_clean_P2, fev1_pp_gli_global_P2, FEV1_FVC_post_P2, and PC1–PC5.

This reduces the sample to n = 1,306. The large drop (4,630 $\rightarrow$ 1,306) is driven primarily by the Phase 3 outcome: 8,912 of 10,652 subjects in pheno_MAL.csv (84%) have no Phase 3 CT scan recorded, meaning they did not return for the follow-up visit ~6 years later.

Table 1: Demographics

Study population after 3-way merge (proteomics $\times$ phenotype $\times$ MAL) and complete-case filtering.

Table 1. Baseline characteristics of COPDGene participants included in the proteomics analysis (n = 1,306). Subjects had complete SomaScan plasma proteomics, phenotype data, and MAL score. Continuous variables: mean (SD); categorical: n (%). MAL (Mechanically Affected Lung) = CT-derived score (range 0.01–0.40); higher values indicate more lung under destructive mechanical stress. Emphysema progression = CT lung density at Phase 3 minus Phase 2 ($\rho_{P3} - \rho_{P2}$, HU); negative values indicate worsening.

Characteristic	Overall (n = 1,306)
Demographics	
Age (years)	65 (8)
Sex: male / female	633 (48%) / 673 (52%)
Race: White / Black	991 (76%) / 315 (24%)
Smoking	
Current smoker	445 (34%)
Pack-years	43 (23)
Pulmonary function	
FEV_1% predicted (GLI global)	83 (24)
FEV_1/FVC	0.69 (0.14)
CT imaging	
Lung density Phase 2 (HU)	83 (23)
MAL score	0.04 (0.05)
Emphysema progression (HU)	1 (7)
Anthropometrics	
BMI (kg/m^2)	29 (6)

Note. Sex is coded 1/2 in the data; assumed Male/Female per COPDGene convention. Race coded 1/2; assumed White/Black. Emphysema progression mean of 1 HU (SD 7) reflects a near-zero mean; the distribution includes both progressors (negative $\Delta\rho$) and improvers (positive $\Delta\rho$).

Figure 1: Study Design

Not automatically generated. Requires a manual schematic (e.g., BioRender) illustrating the analysis flow: SomaScan proteomics $\rightarrow$ differential expression $\rightarrow$ GSEA (Gene Set Enrichment Analysis)

/ STRING → adaptive LASSO (Least Absolute Shrinkage and Selection Operator) proRS (Proteomic Risk Score) → emphysema progression association → causal mediation.

[Figure 1 placeholder: study design schematic to be drawn manually, e.g., BioRender]

Figure 1: Overview of the COPDGene MAL omics analysis pipeline. SomaScan plasma proteomics ($\sim$4,979 proteins, Phase 2) are used to identify proteins associated with MAL (differential expression, GSEA, STRING network), build a proteomic risk score (adaptive LASSO), test whether the score predicts emphysema progression (Phase 3 minus Phase 2 CT density), and quantify mediation of the MAL effect through blood proteins (causal mediation analysis). Schematic to be drawn manually.

Figure 2: Protein Expression Landscape of MAL

A. Volcano Plot (protein $\sim$ MAL)

Each dot is one protein. The x-axis ($\hat{\beta}$) shows how much a protein changes per one-unit increase in MAL after adjusting for age, sex, race, smoking, pack-years, scanner, and ancestry PCs. Positive $\hat{\beta}$ = protein is higher with more mechanical lung stress; negative = lower. $\hat{\beta}$ is defined as the change in $\log_2$ protein per *one-unit* increase in MAL. But MAL only ranges from 0.01 to 0.40 in this cohort, so a one-unit change never actually occurs. To express the effect in realistic terms: multiplying $\hat{\beta}$ by the full MAL range ($0.40 - 0.01 = 0.39$) gives the maximum expected protein change across the entire observed spread of MAL. For Adiponectin: $0.27 \times 0.39 \approx 0.1$ $\log_2$-unit shift from the lowest to the highest MAL subject. The y-axis ($-\log_{10}$ FDR [false discovery rate]) is confidence: higher = more significant. Dots above the threshold (FDR < 0.05) are coloured red (enriched) or blue (depleted); grey dots did not reach significance. A dot far right and high = reliably elevated with more MAL. A dot far left and high = reliably lower with more MAL.

How calculated. For each of 4,979 proteins:

$$\log_2(\text{protein}) \;\sim\; \beta_{\text{MAL}} \cdot \texttt{meanmal} \;+\; \text{age} + \text{sex} + \text{race} + \text{pack-years} + \text{smoking} + \text{scanner} + \text{PC1–PC5}$$

$\hat{\beta}_{\text{MAL}}$ = change in $\log_2$ protein level per one-unit increase in MAL. FDR correction (Benjamini–Hochberg) across all 4,979 tests.

Key result. 45 of 4,979 proteins significant at FDR < 0.05.

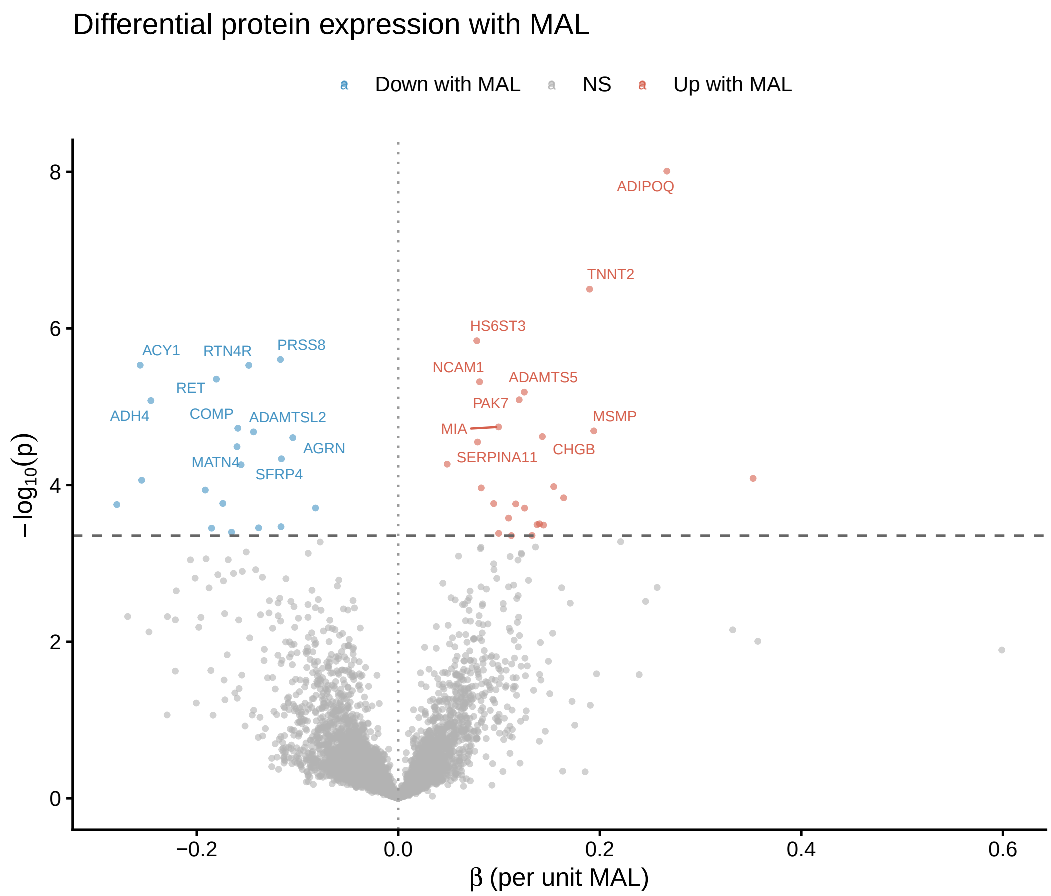


Figure 2: Differential plasma protein expression associated with Mechanically Affected Lung (MAL) score in COPDGene participants (n = 1,306; 4,979 SomaScan proteins). Each point is one protein. x-axis: regression coefficient ($\hat{\beta}$) for $\log_2$ protein level per unit MAL, adjusted for age, sex, race, pack-years, smoking, CT scanner, and ancestry PCs (PC1–PC5). y-axis: $-\log_{10}$(FDR; Benjamini–Hochberg). Red = FDR < 0.05, positively associated; blue = FDR < 0.05, negatively associated; grey = not significant. Top 20 proteins by FDR labelled. 45 proteins reached FDR < 0.05.

Top enriched (higher with more MAL): *ADIPOQ* (Adiponectin, $\beta = 0.27$), *TNNT2* (Troponin T, $\beta = 0.19$), *ADAMTS5*, *NCAM1*, *PAK7*.

Top depleted (lower with more MAL): *ACY1* ($\beta = -0.26$), *ADH4* ($\beta = -0.25$), *RTN4R* (Nogo Receptor), *RET*, *PRSS8*, *SFRP4*.

A2. Sensitivity Analysis: Adjusting for CAD and CHF

To assess whether MAL-associated protein signals reflect cardiovascular comorbidity rather than mechanical lung injury per se, the differential expression analysis was repeated with two additional covariates: history of coronary artery disease (CAD; `CoronaryArtery_P2`, binary) and history of congestive heart failure (CHF; `congestheartfail_slv_P2`, self-reported validated, binary), both from `CellData.csv` (Phase 2 visit). All n = 1,306 analysis subjects had complete CAD and CHF data; no subjects were excluded. CAD prevalence was 7.5% (n = 99); CHF prevalence was 1.2% (n = 16).

The model for each protein was:

$$\log_2(\text{protein}) \sim \text{MAL+age+sex+race+pack-years+smoking+BMI+scanner+PC1–PC5+CAD+CHF}$$

Results are saved to `mal_de_sensitivity_cad_chf.csv`. Proteins that remain FDR < 0.05 after CAD/CHF adjustment are considered robust to cardiovascular comorbidity confounding.

B. GSEA Barplot

GSEA asks whether an entire biological pathway shows a coordinated shift with MAL, rather than asking protein by protein. Each bar is one significant pathway. Bar pointing right (NES > 0): the pathway's proteins are on average higher in plasma with higher MAL. Bar pointing left (NES < 0): lower. Longer bar = stronger shift. NES of ± 2 is a strong, consistent signal. padj is the Benjamini–Hochberg adjusted p-value across all pathways tested; only bars with padj < 0.05 are shown.

How calculated. All 4,979 proteins ranked by their MAL t-statistic (most positively associated at top, most negatively at bottom). Gene sets tested: Hallmark, KEGG Legacy, Reactome (gene-set sizes 15–500).

NES (Normalized Enrichment Score): measures whether a gene set's members concentrate at the top or bottom of the ranked list. NES > 0 = enriched in high-MAL subjects; NES < 0 = depleted. Normalization accounts for gene-set size so scores are comparable across pathways.

padj (adjusted p-value): fgsea tests hundreds of pathways at once. Without correction, many would appear significant by chance. padj is the Benjamini–Hochberg correction applied within each gene set collection (Hallmark, KEGG, Reactome separately). A padj < 0.05 means fewer than 5% of pathways at this threshold are expected to be false positives. Only pathways with padj < 0.05 are shown in the barplot.

Results. 9 pathways significant (padj < 0.05). Seven are depleted (NES < 0): oxidative phosphorylation, glycolysis, fatty acid metabolism, amino acid handling, and xenobiotic processing. Two are enriched (NES > 0): chemokine receptor signalling and antimicrobial peptides (both inflammatory/immune). Higher MAL is associated with lower circulating metabolic proteins and higher inflammatory proteins.

Gene Set Enrichment Analysis (GSEA): significant pathways associated with MAL. Proteins ranked by MAL t-statistic ($n = 4{,}979$) and tested using `fgsea` against Hallmark, KEGG Legacy, and Reactome gene sets (sizes 15–500). NES = Normalized Enrichment Score; NES > 0 = pathway proteins higher in high-MAL subjects (enriched); NES < 0 = lower (depleted). padj = Benjamini–Hochberg adjusted p-value within each collection. Only pathways with padj < 0.05 shown ($n = 9$).

Collection	Pathway	NES	padj
Hallmark	Fatty Acid Metabolism	−1.76	0.021
Hallmark	Oxidative Phosphorylation	−1.78	0.021
KEGG	Glycolysis / Gluconeogenesis	−2.02	0.018
Reactome	Biological Oxidations	−2.05	0.007
Reactome	Metabolism of Amino Acids & Derivatives	−1.84	0.013
Reactome	Diseases of Metabolism	−1.76	0.032
Reactome	Phase I Functionalization of Compounds	−1.97	0.037
Reactome	Chemokine Receptors Bind Chemokines	+2.02	0.024
Reactome	Antimicrobial Peptides	+1.94	0.032

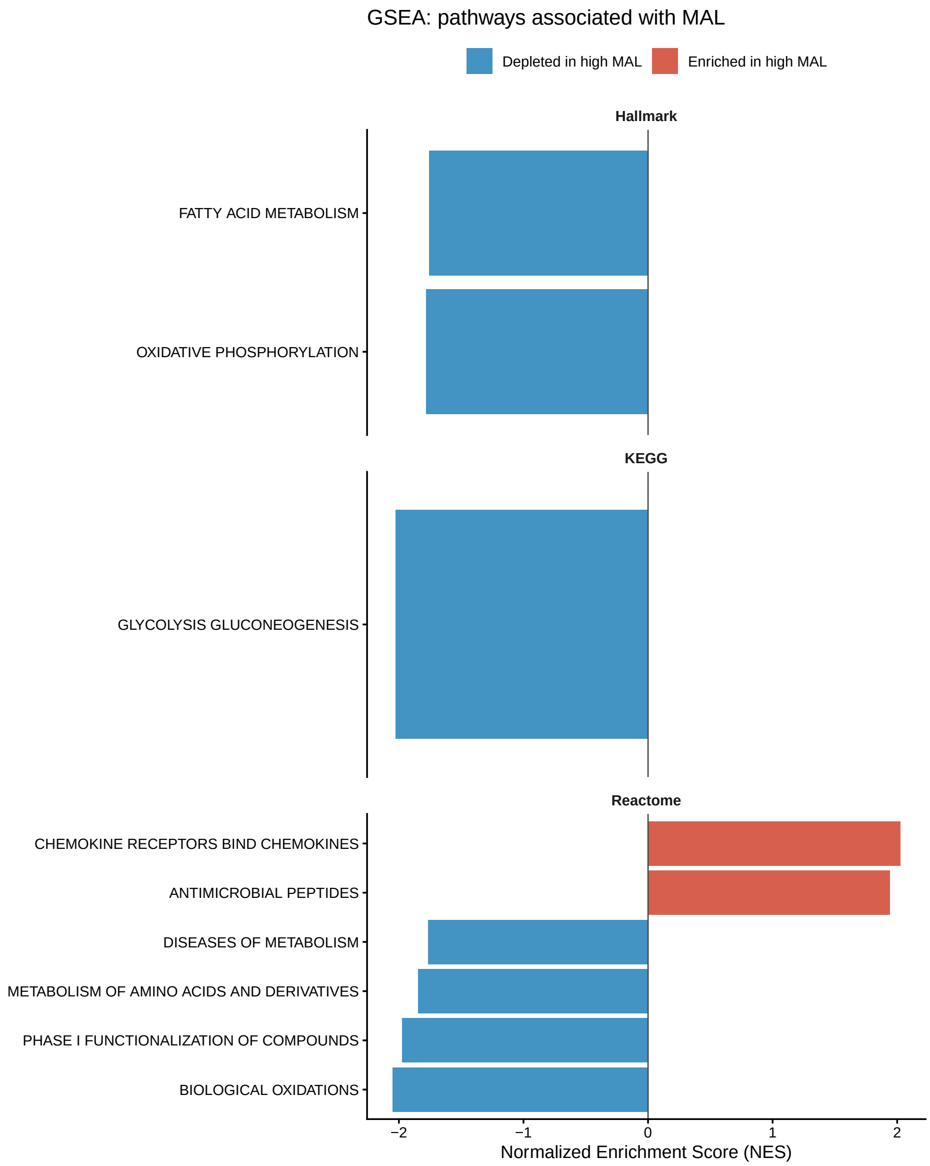


Figure 3: Gene Set Enrichment Analysis (GSEA) of MAL-associated plasma proteins across Hallmark, KEGG Legacy, and Reactome collections. Proteins ranked by MAL t-statistic ($n = 4{,}979$); bar length represents the Normalized Enrichment Score (NES). NES > 0 = enriched in high-MAL subjects; NES < 0 = depleted. Colour indicates direction. Only pathways with Benjamini–Hochberg adjusted $p < 0.05$ are shown ($n = 9$; 7 depleted, 2 enriched).

C. STRING Network

Protein–protein interaction network for the 45 MAL-associated proteins. Each node is one protein. An edge between two proteins means they are known to physically interact or co-function in humans per STRING. A cluster of connected nodes indicates those proteins share a biological process. A hub (many edges) may be a central regulator in the MAL response. An isolated node had no high-confidence interactions with the other significant proteins. Edge confidence ≥ 400 (medium confidence).

How calculated. 45 FDR-significant proteins submitted to STRING v11.5 human database. Interactions with confidence ≥ 400 drawn. Only PNG available: `STRINGdb::plot_network()` does not support PDF output.

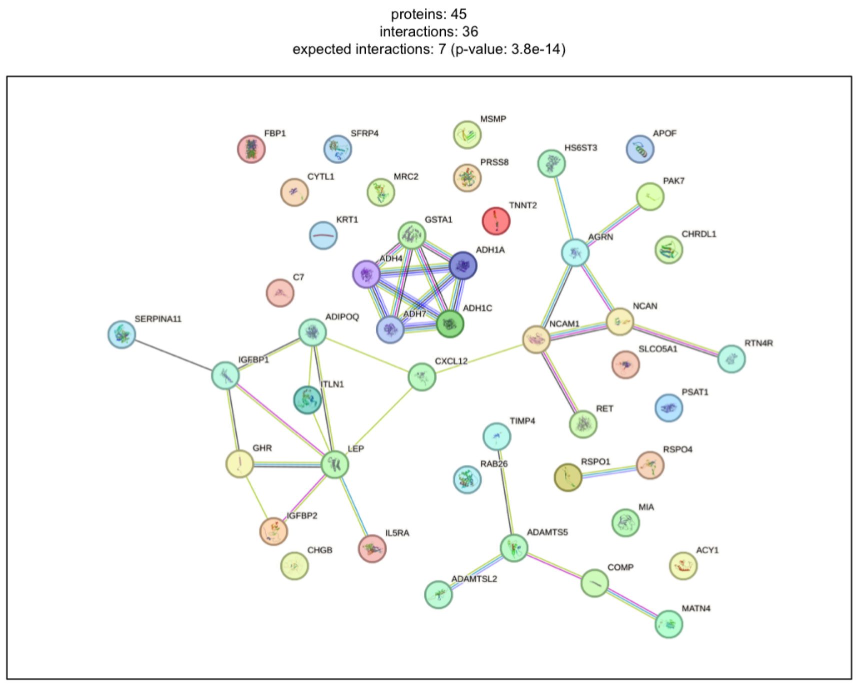


Figure 4: Protein–protein interaction network for the 45 plasma proteins significantly associated with MAL (FDR < 0.05), constructed using STRING v11.5 (human, species 9606). Nodes represent proteins; edges indicate known or predicted functional interactions with confidence score ≥ 400 (medium confidence). Isolated nodes had no qualifying interactions within this set. Network image exported as PNG; PDF output is not supported by `STRINGdb::plot_network()`.

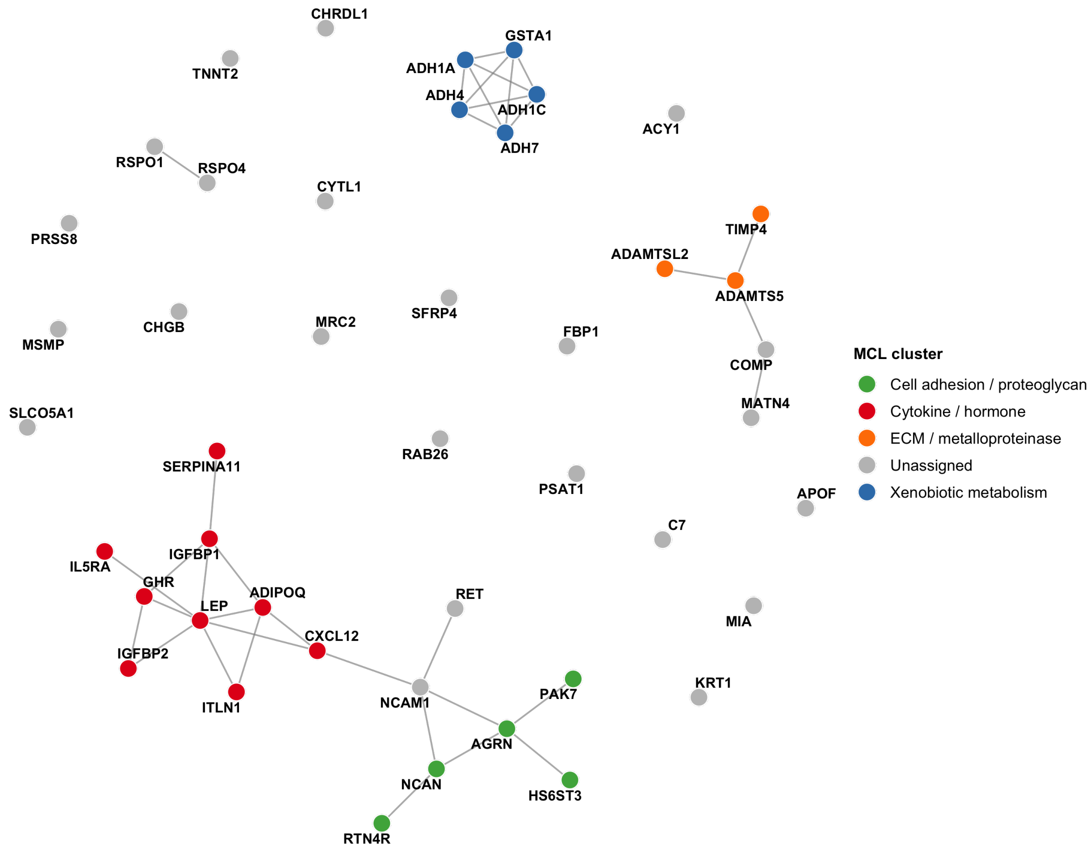
C2. MCL Clustering and Cluster Enrichment

Markov Cluster Algorithm (MCL) clustering (inflation parameter = 2, default) was applied to the STRING interaction network to partition the 45 MAL-associated proteins into biologically coherent modules. MCL was run using the `MCL` R package (`mcl()`, `addLoops=TRUE`, `inflation=2`) on the binary adjacency matrix of the simplified subnetwork. Proteins not assigned to any cluster (cluster 0, $n = 17$) were excluded. Clusters with fewer than 3 proteins were also excluded from enrichment analysis. For each retained cluster, pathway enrichment was retrieved from the STRING API across Gene Ontology (GO:BP, GO:MF, GO:CC), KEGG, and Reactome databases; FDR was computed by STRING using the Benjamini–Hochberg procedure against the human proteome background. Full results are in `STRING_mcl_clusters.csv` and `STRING_cluster_enrichment.csv`.

Four clusters with ≥ 3 proteins emerged (covering 22 of 45 proteins assigned; 17 unassigned): a cytokine/hormone signaling cluster ($n = 9$), a xenobiotic/alcohol metabolism cluster ($n = 5$), a cell adhesion and heparan sulfate proteoglycan cluster ($n = 5$), and an ADAMTS metalloproteinase cluster ($n = 3$).

MCL cluster enrichment summary. Top enriched pathway per cluster (lowest FDR across GO:BP, KEGG, and Reactome). Clusters with ≥ 3 proteins are shown. KEGG = Kyoto Encyclopedia of Genes and Genomes; RCTM = Reactome; GO:CC = Gene Ontology Cellular Component. Full results in `STRING_cluster_enrichment.csv`.

Cluster	n	Members	Top pathway	DB	FDR
1	9	ADIPOQ, CXCL12, GHR, IGFBP1, IGFBP2, IL5RA, ITLN1, LEP, SERPINA11	Cytokine-cytokine receptor interaction	KEGG	0.0018
2	5	ADH1A, ADH1C, ADH4, ADH7, GSTA1	Metabolism of xenobiotics by cytochrome P450	KEGG	1.6×10^{-10}
3	5	AGRN, HS6ST3, NCAN, PAK7, RTN4R	Heparan sulfate/heparin (HS-GAG) metabolism	RCTM	5.3×10^{-4}
4	3	ADAMTS5, ADAMTSL2, TIMP4	Extracellular matrix	GO:CC	0.034



STRING network with MCL cluster assignments. Nodes represent the 45 MAL-associated proteins (FDR < 0.05); edges represent STRING interactions (combined score ≥ 400). Node colours indicate MCL cluster membership (inflation $= 2$). Four clusters with ≥ 3 proteins are highlighted; 17 proteins not assigned to any cluster are shown in grey.

Proteomic Risk Score: Methods

A progression-targeting Proteomic Risk Score (proRS) was built using adaptive elastic net ($\alpha = 0.5$) with emphysema progression (ΔCT density, HU) as the direct outcome. SomaScan RFU values were $\log_2$-transformed; a 70/30 train/test split was applied (`set.seed(2024)`, $n_{\text{train}} = 914$, $n_{\text{test}} = 392$). Ridge regression on residualised progression (after projecting out age, sex, race, smoking status, pack-years, and CT scanner type) produced adaptive penalty weights $w_j = 1/|\hat{\beta}_j^{\text{ridge}}|$. Adaptive elastic net with 10-fold cross-validation ($\lambda_{\min}$) selected 410 proteins. The proRS is the weighted sum of selected $\log_2$-transformed proteins, standardised to mean 0, SD 1 using the training set mean and SD (applied to all subjects). Higher proRS = greater predicted CT density loss (more emphysema worsening). Test $R^2 = 0.039$: this model selects proteins with effects *independent of* the clinical covariates and is used for the protein annotation tables and biological interpretation.

A separate *prognostic* elastic net (proteins only, no unpenalized clinical covariates; same adaptive elastic net framework, $\alpha = 0.5$, 10-fold inner CV) was fitted on the training set to produce a proRS for the AIC comparison and Figure 3. Training without clinical covariates ensures that the proRS captures total protein-derived signal, enabling a proper test of whether proteins add beyond clinical factors. This model selected 380 proteins (test $R^2 = 0.021$).

For Figure 3, the association model, and the AIC comparison, 5-fold cross-validated out-of-sample predictions from the prognostic model are used: the full cohort (n = 1,306) is partitioned into 5 folds; each fold is predicted by a prognostic elastic net fitted on the remaining four folds, so every subject's proRS is an out-of-sample prediction. The full-cohort CV predictions are standardised to mean 0, SD 1 on their own distribution (all OOS, no leakage). Adaptive weights and lambda are recomputed within each outer fold using 5-fold inner CV.

Note: the MAL adaptive LASSO proRS (378 proteins, from `MAL_omics_proteomics.ipynb`) characterises blood protein patterns that reflect current mechanical lung stress and is used for the causal mediation analysis (Table 2). The progression-targeting elastic net proRS is used for Figure 3 and the AIC comparison.

Figure 3: Emphysema Worsening by Proteomic Risk Score Quartile

Subjects are divided into four quartiles by the Proteomic Risk Score (proRS). Q1 = lowest-risk 25% (lowest proRS); Q4 = highest-risk 25% (highest proRS). The y-axis shows the OLS $\hat{\beta}$ coefficient for each quartile indicator versus Q1 (reference), from the model:

$$\Delta\text{CT density} \sim \text{factor(quartile)} + \text{age} + \text{sex} + \text{race} + \text{smoking} + \text{pack-years} + \text{BMI} + \text{scanner}$$

Each $\hat{\beta}$ is the covariate-adjusted mean difference in emphysema progression relative to Q1; positive = more CT density loss than Q1. Error bars are 95% CI.

Key result. Q2: $\hat{\beta} = 0.95$ HU (95% CI $[-0.12,\ 2.03]$, $p = 0.08$); Q3: $\hat{\beta} = 1.41$ HU ($[0.31,\ 2.51]$, $p = 0.01$); Q4: $\hat{\beta} = 1.92$ HU ($[0.81,\ 3.04]$, $p < 0.001$). A monotone gradient is present; Q3 and Q4 reach statistical significance, with subjects in the highest proRS quartile showing 1.9 HU more emphysema worsening than Q1 after covariate adjustment.

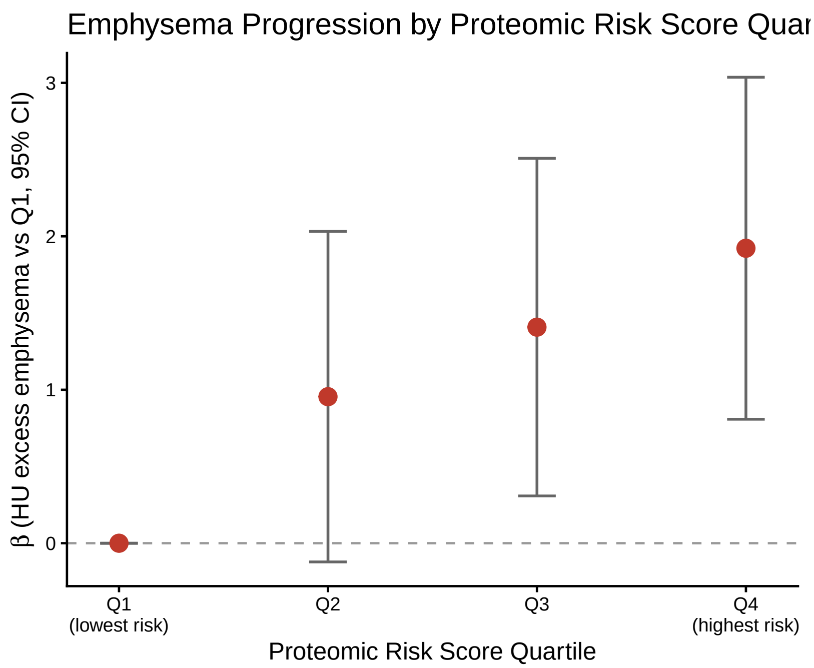


Figure 5: $\hat{\beta}$ coefficients (HU excess emphysema worsening relative to Q1, 95% CI) for proRS quartiles in a single OLS model: ΔCT density $\sim$ factor(quartile) + age + sex + race + smoking + pack-years + BMI + scanner, n = 1,306. Subjects ranked into quartiles by 5-fold cross-validated out-of-sample predictions from the prognostic elastic net (main model: 380 proteins, $\alpha = 0.5$, test $R^2 = 0.021$). Q1 = lowest-risk 25% (reference, $\hat{\beta} = 0$); Q4 = highest-risk 25%. Positive $\hat{\beta}$ = more CT density loss than Q1. Error bars are 95% CI.

Association of Proteomic Risk Score with Emphysema Progression

The proRS was tested as a predictor of emphysema progression in a multivariable linear regression model adjusting for age, sex, race, current smoking status, pack-years of smoking, BMI, and CT scanner type:

$$\Delta\text{CT density} \sim \text{proRS} + \text{age} + \text{sex} + \text{race} + \text{smoking} + \text{pack-years} + \text{BMI} + \text{scanner}$$

Proteomic Risk Score $\hat{\beta} = 0.53$ HU/SD, 95% CI $[0.13,\ 0.93]$, $p = 0.009$ (5-fold CV OOS, n = 1,306).

Each SD increase in the proRS is associated with 0.53 HU greater CT density loss (more emphysema worsening) after adjustment for age, sex, race, smoking, pack-years, BMI, and CT scanner type.

AIC Comparison: Proteomic Risk Score vs. Clinical Model

Three models predicting emphysema progression are compared by AIC (Akaike Information Criterion: $-2\log\hat{L} + 2k$; lower = better). The proRS values used here are 5-fold cross-validated out-of-sample predictions: the full cohort (n = 1,306) is split into 5 folds; each fold is predicted by an elastic net trained on the other four folds, so every subject's proRS is an out-of-sample prediction. This approach eliminates in-sample inflation while using the full cohort for maximum statistical power. ΔAIC relative to the clinical reference; $\Delta \leq -2$ indicates meaningful improvement.

- **Clinical:** age, sex, race, current smoking, pack-years, BMI. Reference model.
- **Proteomic Risk Score:** 5-fold CV elastic net proRS (380 proteins, $\alpha = 0.5$, proteins only) alone.
- **Clinical + Proteomic Risk Score:** clinical variables plus CV proRS.

Model comparison for predicting emphysema progression by AIC (full cohort 5-fold CV, n = 1,306; lower = better fit). Proteomic Risk Score = 5-fold cross-validated OOS predictions from the prognostic elastic net (380 proteins, proteins only, $\alpha = 0.5$, test $R^2 = 0.021$). BMI from `CellData.csv` (Phase 2 visit). ΔAIC relative to clinical reference; $\Delta \leq -2$ indicates meaningful improvement.

Model	AIC	ΔAIC
Clinical (age, sex, race, smoking, pack-years, BMI)	8,900.5	Reference
Proteomic Risk Score alone	8,920.0	+19.1
Clinical + Proteomic Risk Score	8,886.4	−14.1

proRS $\hat{\beta} = 0.84$ HU/SD, 95% CI [0.43, 1.24], $p < 0.001$ (in the Clinical + Proteomic Risk Score model).

Key result. The proteomic risk score alone does not outperform the clinical reference (ΔAIC = +19.1), consistent with proteins lacking the direct demographic and behavioural context captured by age, sex, race, and smoking history. However, adding the proRS to the clinical model yields a meaningful improvement (ΔAIC = −14.1, exceeding the ≤ -2 threshold; $\hat{\beta} = 0.84$ HU/SD, $p < 0.001$), demonstrating that blood protein patterns capture emphysema progression risk beyond standard clinical predictors. The out-of-sample metric is test $R^2 = 0.021$ and a Q1-to-Q4 gradient of 1.9 HU (Figure 3).

Table 2: Causal Mediation

The analysis decomposes the total effect of MAL on emphysema progression into a direct path (NDE) and a path through the blood protein score (NIE). NDE = how much of the MAL effect goes through pathways other than the protein score. NIE = how much goes through the protein score. All effects are in HU (Hounsfield Units), the CT density scale. More negative HU = lower lung density = more emphysema. A difference of 1 HU is a small change; the SD of progression in this cohort is ∼7 HU.

- **NDE = 19.7 HU,** $p = 0.004$**:** MAL drives progression through mechanisms independent of the protein score. A one-unit increase in MAL is associated with 19.7 HU more progression via the direct path.
- **NIE =** -0.12 **HU,** $p = 0.97$**:** essentially zero and non-significant. The proteins are not the mechanism by which MAL causes progression.
- **Proportion mediated =** -0.6%**:** of the total MAL effect, -0.6% is explained by the blood protein pathway. Negative proportion arises when the indirect effect goes in the opposite direction to the total effect; here it reflects noise around zero.
- **Total effect = 19.6 HU:** NDE + NIE, almost entirely direct.

How calculated. Pathway tested: MAL → Proteomic Risk Score → Emphysema Progression. Fitted using the `medflex` R package (natural effects framework) with robust sandwich standard errors.

Two models fitted internally by `medflex`:

$$\text{Imputation model: } \Delta\rho \sim \texttt{meanmal} \times \text{proRS} + \text{covariates}$$

$$\text{Natural effects model: } \Delta\rho \sim \texttt{meanmal}_0 + \texttt{meanmal}_1 + \text{covariates}$$

$\texttt{meanmal}_0$ is the MAL value at which the mediator (proRS) is set (direct path); $\texttt{meanmal}_1$ is the actual MAL value driving the outcome (indirect path). The difference between them isolates the mediated portion.

Table 2. Causal mediation of the MAL–emphysema progression relationship through the Proteomic Risk Score (proRS), estimated using the natural effects framework (`medflex` R package, robust sandwich SE, $n = 1{,}306$). NDE (Natural Direct Effect) = portion of MAL's effect on ΔCT lung density not operating through the proRS. NIE (Natural Indirect Effect) = portion operating through the proRS. Proportion mediated = NIE / total effect. All effects in HU (Hounsfield Units; negative = more emphysema).

Effect	Estimate (HU)	z	p
Natural Direct Effect (NDE)	+19.72	2.92	0.004
Natural Indirect Effect (NIE)	−0.12	−0.04	0.971
Total effect	+19.60	N/A	N/A
Proportion mediated	−0.6%		

Key result. MAL has a large, significant direct effect on emphysema progression (NDE $p = 0.004$). The proteomic risk score mediates essentially none of this effect (NIE $p = 0.97$, proportion $= -0.6\%$).

Supplement

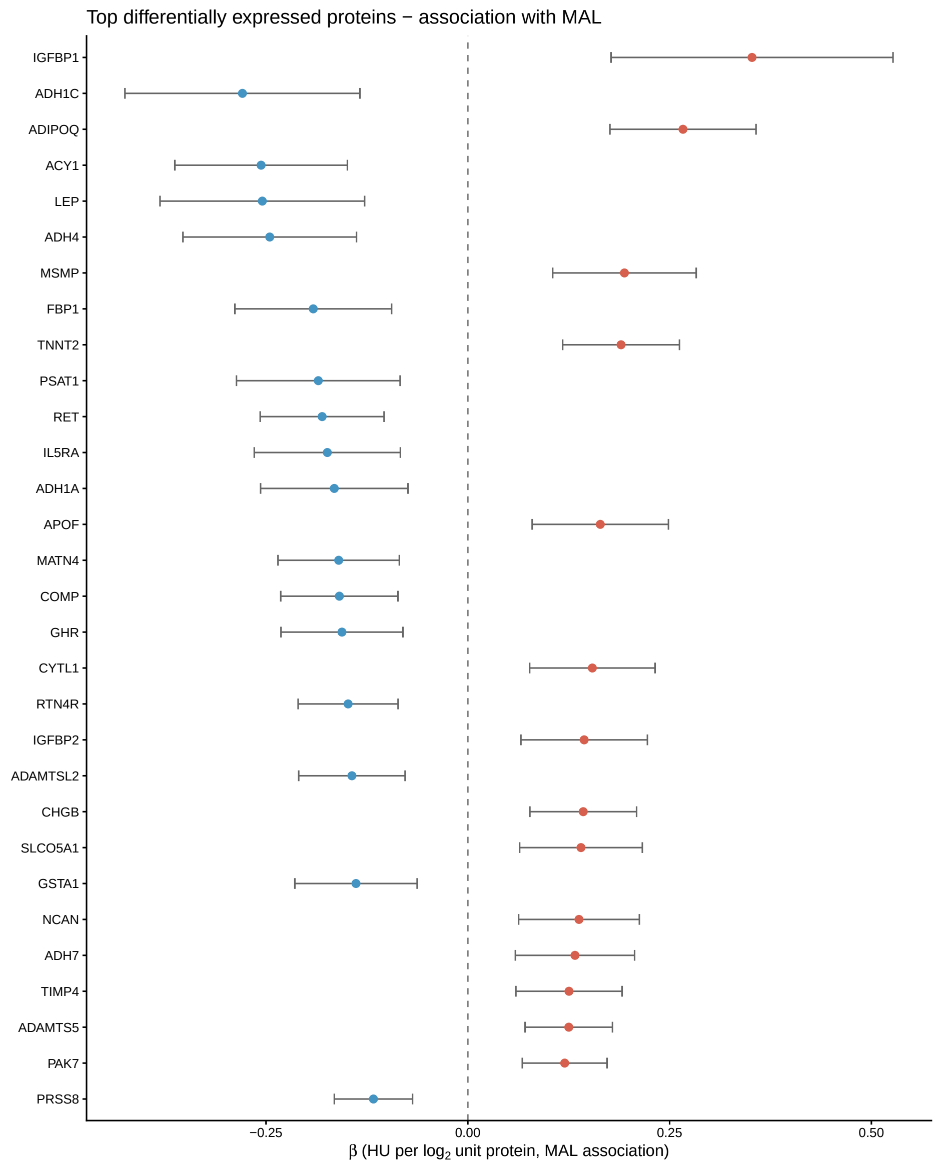
Supplementary Tables

- **S1: All differential expression results** (4,979 proteins): gene, target, $\hat{\beta}$, SE, t, p, FDR.
  File: `MAL_omics/mal_de_results.csv`
- **S2: proRS weights** (MAL-targeting LASSO proteins): gene, target, $\hat{\beta}_{\text{OLS}}$, 95% CI, p.
  File: `MAL_omics/mal_risk_score_weights.csv`
- **S3: Per-protein mediation** (MAL-targeting LASSO proteins): NDE, NIE, total, proportion mediated, NIE p.
  File: `MAL_omics/mal_mediation_supp.csv`

  Top 6 individual mediators (NIE $p < 0.05$): TREM2 (13%, $p = 0.017$), ADAMTS5 (5.7%, $p = 0.021$), SEZ6L (17%, $p = 0.025$), RSPO1 (14%, $p = 0.038$), GREM2 (12%, $p = 0.039$), ADH7 (7%, $p = 0.044$). All small in absolute magnitude.

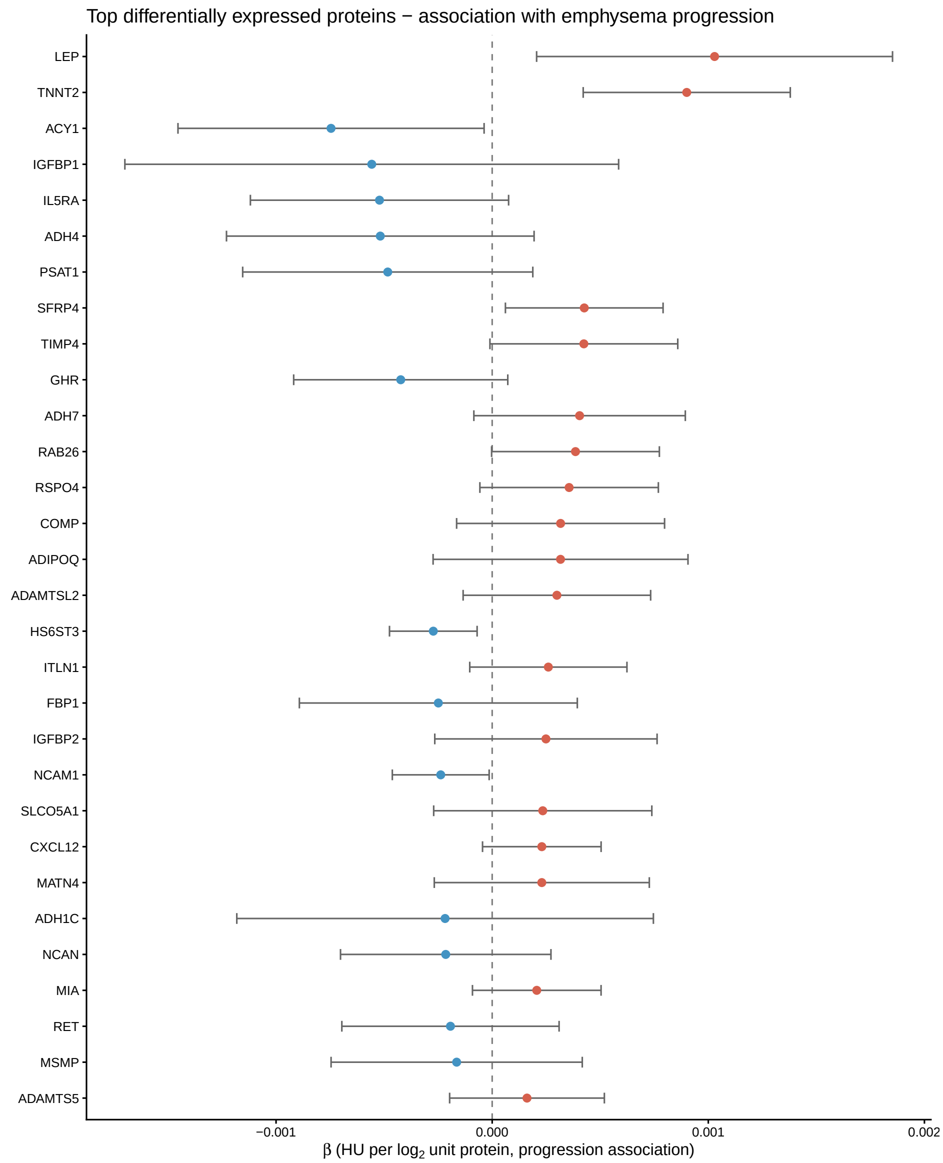
Supplementary Figures

- **Supp Figure A: Forest plot, proteins $\sim$ MAL.** Top 30 FDR-significant proteins, $\hat{\beta}$ for MAL association with 95% CI.



Supplementary Figure A. Forest plot of the top 30 proteins most significantly associated with MAL score (ranked by FDR). Points show OLS $\hat{\beta}$ ($\log_2$ RFU per unit MAL) with 95% CI, from the model: $\log_2(\text{protein}) \sim \texttt{meanmal} + \text{age} + \text{sex} + \text{race} + \text{pack-years} + \text{smoking} + \text{scanner} + \text{PC1–PC5}$. Red = positively associated; blue = negatively associated with MAL. Vertical dashed line at zero.

- **Supp Figure B: Forest plot, proteins $\sim$ emphysema progression.** Same 30 proteins, $\hat{\beta}$ for progression association (adjusted for baseline lung density and covariates).



Supplementary Figure B. Forest plot of the same top 30 MAL-associated proteins (ordered as in Supplementary Figure A) showing their association with emphysema progression (ΔCT lung density, HU). Points show OLS $\hat{\beta}$ with 95% CI, adjusted for age, sex, race, pack-years, smoking, BMI, and scanner. Vertical dashed line at zero.

Assumptions Made

1. **Figure 1 (study schematic)**: not automatically generated. Needs to be drawn manually.
2. **BMI**: sourced from `CellData.csv` (column `BMI`, filtered to `Phase_study == 2`). All 1,306 analysis subjects have Phase 2 BMI (mean 29, SD 6). Included in Table 1 and the clinical model.
3. **FEV$_1$% predicted**: column `fev1_pp_gli_global_P2` (GLI global reference equation) confirmed present in `pheno_MAL.csv`. Mean 83 (SD 24) in Table 1.